Supplementary Material for
Joint Model Selection and Adaptation for Few-Shot Predictive-Model Instantiation in Digital Twin Platforms under Deployment Limits

This supplementary material provides implementation, analytical, and diagnostic details referenced in the main manuscript. It includes the operation-count and parameter-count rules, candidate-bank records, relative-margin calibration, additional analytical results, target-side baseline settings, diagnostic comparisons, hardware profiling, configuration analysis, external-transfer results, and synthetic-benchmark generation details.

The source, architecture-screening, margin-calibration, diagnostic, and held-out target pools use disjoint center identities.

S1 Detailed Operation-Count and Parameter-Count Rules

The estimated operation count is computed for one forward pass with batch size one. Additions and multiplications in the main linear, convolutional, recurrent-matrix, residual-projection, and output-layer operations are counted. Bias additions, nonlinear activations, dropout operations, and GRU elementwise operations are excluded. The same deterministic rule is applied to every candidate.

For an instantiated architecture a_q, the parameter count is the exact number of elements in all trainable parameter tensors after the target input dimension and prediction horizon are fixed. Both counts depend on the target case only through its instantiated input and output dimensions. Parameter adaptation changes neither count.

The two indicators are used as hard deployment limits. They are not added to the prediction loss and are not changed after target adaptation. They are deterministic model-level indicators rather than direct guarantees of hardware-specific latency or memory use. Their relation to measured inference latency and serialized model size is evaluated in Supplementary Section S5.

S2 Candidate-Bank Records and Relative-Margin Calibration

S2.1 Candidate-bank construction and frozen records

The initial architecture space contains 66 indexed configurations: 18 MLPs, 36 TCNs, and 12 GRUs. The pooled-source PT-A57 initialization, a one-layer GRU with 32 hidden units, is fixed as the protected reference candidate before screening. Candidate discovery separately evaluates source-trained C1 candidates for all 66 indexed configurations, including a C1-A57 initialization distinct from PT-A57.

A C1 configuration is retained if it is the check-loss winner over PT-A57 in at least two screening cases, or if validation selects it and it then records a positive check-loss gain over PT-A57 in at least two cases. These cases belong to the architecture-screening pool and are disjoint from the margin-calibration and held-out target pools. The retained list contains three MLP and three GRU configurations. No TCN configuration satisfies this rule.

Two retained one-layer GRU-32 configurations have different indexed settings, but recurrent dropout is inactive for a one-layer GRU. They therefore represent the same executable architecture. The six retained configurations correspond to five executable architectures.

The final bank contains the protected PT-A57 reference and one source-trained C1 initialization for each of the six retained indexed configurations. The retained C1-A57 candidate is therefore a paired-initialization control rather than the reference. The six retained configurations plus PT-A57 produce seven candidates. The bank and reference candidate are frozen before relative-margin calibration and held-out evaluation.

S2.2 Relative-margin calibration formulation

Let $\mathcal{T} = \{0.05, 0.075, 0.10, 0.125, 0.15, 0.20\}$ denote the preset margin grid. For a relative margin τ, let $\widehat{R}_{\mathrm{harm}}(\tau)$ denote the observed harmful-selection rate. Let $\widehat{G}_{\mathrm{ref}}(\tau)$ and $\widehat{G}_{\mathrm{pair}}(\tau)$ denote the mean check-loss gains over the adapted reference and the paired-initialization control. Their lower 95% confidence bounds are $\underline{G}_{\mathrm{ref}}(\tau)$ and $\underline{G}_{\mathrm{pair}}(\tau)$.

The eligible margins form

$$
\begin{aligned}
\mathcal{T}_{\mathrm{elig}} = \bigl\{\tau \in \mathcal{T} \;\bigm|\; & \widehat{R}_{\mathrm{harm}}(\tau) \le \rho, \\
& \widehat{G}_{\mathrm{ref}}(\tau) \ge \gamma_{\mathrm{ref}}, \quad \underline{G}_{\mathrm{ref}}(\tau) > 0, \\
& \widehat{G}_{\mathrm{pair}}(\tau) \ge \gamma_{\mathrm{pair}}, \quad \underline{G}_{\mathrm{pair}}(\tau) > 0\bigr\}.
\end{aligned}
\tag{S1}
$$

The calibration uses $\rho = 0.05$ and $\gamma_{\mathrm{ref}} = \gamma_{\mathrm{pair}} = 0.03$. When $\mathcal{T}_{\mathrm{elig}}$ is nonempty, the calibrated margin is

$$
\tau^* = \min \mathcal{T}_{\mathrm{elig}}. \tag{S2}
$$

The smallest eligible value in the main margin-calibration setting is $\tau^* = 0.10$.

Architecture screening and candidate-bank construction are completed before margin calibration. The candidate bank and reference candidate are fixed when the margin grid is evaluated.

If no margin is eligible in a new domain, no domain-specific calibrated margin is claimed. In calibrated deployment, the adapted reference is retained and the domain is marked as unsupported for margin calibration. A previously frozen margin can still be evaluated without recalibration as a transfer audit, but it is not treated as a calibrated deployment rule for the new domain. The released calibration CSV contains the complete six-margin grid, harmful-selection rates, mean gains, and confidence bounds.

Table S1 reports the complete grid used to freeze $\tau = 0.10$.

Table S1: Complete relative-margin calibration grid.

τ	Harmful selections	$\widehat{G}_{\mathrm{ref}}$ (95% CI)	$\widehat{G}_{\mathrm{pair}}$ (95% CI)	Eligible
0.050	5/80 (6.25%)	13.93% $[7.75, 20.94]$	13.33% $[7.73, 19.81]$	No
0.075	5/80 (6.25%)	13.79% $[7.77, 20.92]$	13.18% $[7.39, 19.49]$	No
0.100	4/80 (5.00%)	13.85% $[8.06, 20.58]$	13.32% $[7.68, 19.48]$	Yes
0.125	4/80 (5.00%)	13.13% $[7.01, 20.17]$	12.60% $[6.85, 18.91]$	Yes
0.150	4/80 (5.00%)	12.70% $[6.48, 19.75]$	12.17% $[6.30, 18.91]$	Yes
0.200	2/80 (2.50%)	10.98% $[5.56, 17.31]$	10.45% $[5.70, 15.85]$	Yes

The calibration pool contains 20 centers and four cases per center ($N = 80$ cases; 20 independent resampling clusters). Gains are paired check-MSE reductions. Intervals are 95% center-cluster percentile bootstrap intervals with 4000 repetitions. Eligibility applies all criteria in Eq. (S1); the smallest eligible margin is $\tau = 0.10$.

S3 Additional Analytical Details

S3.1 Ideal target decision

Let $R_c(q)$ denote the expected loss of an adapted candidate on future observations from target center c. If this quantity were available, the ideal feasible candidate would satisfy

$$q_c^\dagger \in \arg\min_{q \in \mathcal{Q}_c^{\mathrm{fea}}} R_c(q). \tag{S3}$$

The quantity $R_c(q)$ is not available during model instantiation. MSA-DTI therefore uses the observed validation loss $J_c(q)$ together with the relative-margin rule in the main manuscript.

S3.2 Conditional expected-loss result

Assume that validation loss differs from expected future loss by at most $\epsilon_c \geq 0$ for every feasible candidate:

$$|R_c(q) - J_c(q)| \leq \epsilon_c, \qquad \forall q \in \mathcal{Q}_c^{\mathrm{fea}}. \tag{S4}$$

When MSA-DTI selects an alternative, the relative-margin rule gives

$$R_c(q_c^*) - R_c(q^{\mathrm{ref}}) \leq -\tau J_c(q^{\mathrm{ref}}) + 2\epsilon_c. \tag{S5}$$

Hence, the selected alternative has lower expected future loss than the reference when

$$\tau J_c(q^{\mathrm{ref}}) > 2\epsilon_c. \tag{S6}$$

This is a sufficient condition and is not a guarantee for every target center.

S3.3 Target-stage computational complexity

Let C_q^{grad} be the cost of one target update for candidate q, and let C_q^{val} be its validation cost. The target-stage computational complexity is

$$O\left(|\mathcal{Q}| + \sum_{q \in \mathcal{Q}_c^{\mathrm{fea}}} \left(T C_q^{\mathrm{grad}} + C_q^{\mathrm{val}}\right)\right). \tag{S7}$$

The first term represents deployment filtering. The sum represents target adaptation and validation of the feasible candidates. Offline candidate-bank construction and relative-margin calibration are excluded.

S4 Additional Experimental and Diagnostic Results

S4.1 Target-side baseline settings

The main manuscript keeps only the baseline information needed for the main comparison. Table S2 gives the target-side settings used in the main comparison.

Table S2: Target-side settings of MSA-DTI and the baselines in the main comparison.

Method	Structure	Source initialization	Target update	Adapted models	Deployment-limit check
PT+FT	Fixed reference	Pooled-source model	SGD/MSE–50	1	Before output
MeDeT-based adaptation	Fixed reference	Meta-learned initialization	SGD/MSE–50	1	Before output
Random initialization	Fixed reference	Random	SGD/MSE–50	1	Before output
Few-shot NAS	Top-12 shortlist	Method-specific models	Adam/Huber–50	≤ 12	Before evaluation
Zero-shot NAS	Top-ranked model	Method-specific model	None	0	Before evaluation
Zero-shot NAS + fine-tuning	Top-12 shortlist	Method-specific models	Adam/Huber–50	≤ 12	Before evaluation
H-Meta-NAS-based adaptation	Target architecture search	Architecture-indexed meta-initializations	SGD/MSE–50	Multiple candidates	Before search evaluation
MSA-DTI	Candidate bank	Frozen source initializations	SGD/MSE–50	4–7	Before adaptation and output

PT+FT and MSA-DTI use the same source-trained reference and target adaptation procedure. H-Meta-NAS-based adaptation uses the 50-update budget-matched condition in the main comparison. It uses the same 66 architecture definitions, filters infeasible architectures before search, uses source-only first-order meta-initializations, and selects the final architecture using target validation data. Its settings are fixed without held-out test data. All methods use the same target cases, deployment limits, and architecture definitions.

S4.2 Optimizer-matched diagnostic comparison

This comparison examines whether the performance ordering changes after the target optimizer, task loss, and update budget are matched. All methods use the same independent diagnostic target cases and the same deployment limits. Table S3 reports the full optimizer-matched comparison.

Table S3: Full optimizer-matched comparison on the diagnostic target pool.

Method	MSE↓	Worst-10%↓	CVaR90↓	Adapted models
MSA-DTI	**0.003809**	**0.015094**	**0.008696**	6.10
PT+FT	0.004497	0.017144	0.011878	1.00
Few-shot NAS	0.006715	0.025246	0.019653	12.00
Zero-shot NAS + fine-tuning	0.007402	0.028099	0.020402	12.00
Common shortlist, lowest validation loss	0.006707	0.025240	0.019653	12.00
Common shortlist, reference-based selection	0.006819	0.025562	0.020547	12.00

MSA-DTI gives the lowest MSE, Worst-10% error, and CVaR90 in this comparison. The source initializations and selection rules still differ. This comparison is therefore a robustness check rather than a complete method equivalence test.

S4.3 Component variants

The following diagnostic pool is separate from the source, architecture-screening, margin-calibration, and held-out target pools. It is used only for the component comparisons reported in this subsection. Table S4 summarizes these component variants.

Table S4: Component variants and diagnostic comparisons.

Variant	MSE↓	Worst-10%↓	Paired MSE reduction (%)↑	Harmful rate (%)↓	Limit satisfaction (%)↑
MSA-DTI	0.004063	0.015764	15.79	15.00	100.00
Earlier source-training procedure for alternative models	0.004876	0.019044	1.75	2.50	100.00
Reference candidate only	0.005051	0.019512	0.00	0.00	100.00
Two reference-architecture initializations	0.004917	0.019136	1.29	1.25	100.00
Without relative margin	0.004055	0.015690	16.16	17.50	100.00
Without deployment filtering	0.004037	0.015635	16.13	13.75	96.25

Paired MSE reduction uses the adapted reference on the same diagnostic pool. Harmful rate is computed over all diagnostic cases. The earlier source-training row changes the complete source-training procedure for the alternative models and is not a single-component ablation.

The historical source-training variant changes the complete procedure used to construct the alternative source initializations and is therefore interpreted as candidate-bank sensitivity rather than the isolated effect of a single training choice. The reference-only and two-reference-initialization variants show that additional initializations of the reference architecture do not explain the gain from cross-architecture selection. Removing deployment filtering slightly lowers diagnostic MSE but reduces limit satisfaction from 100% to 96.25%, confirming the role of filtering in enforcing the hard limits.

S4.4 Selection-outcome details

On the 80 independent diagnostic cases, the relative-margin rule selects alternatives in 61 cases, including 49 beneficial and 12 harmful selections. Direct lowest-validation-loss selection chooses alternatives in 69 cases, including 55 beneficial and 14 harmful selections. The corresponding all-case harmful-selection rates are 15.00% and 17.50%, respectively. Figure S1 summarizes these outcomes.

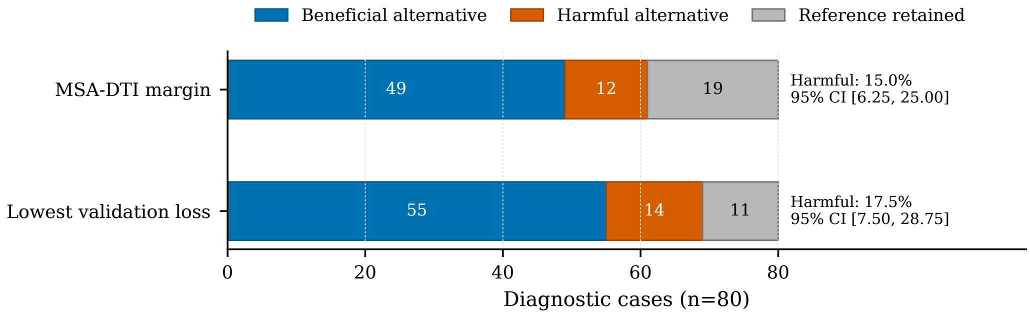

Figure S1: Selection outcomes on the 80-case diagnostic pool.

The relative-margin rule retains the reference more often. The confidence intervals of the two harmful-selection rates overlap, so the observed difference is not statistically confirmed.

S5 Timing, Hardware Profiling, and Deployment Analysis

S5.1 Target-side timing protocol

All methods are implemented in Python and PyTorch and run on an NVIDIA RTX 3060 Laptop GPU. Target-side time includes model creation, initialization loading, target adaptation, validation evaluation, and final selection. It excludes offline preparation and test evaluation. Each reported time is the mean over five runs, and the main manuscript reports the corresponding standard deviation. Before collection, each method receives one unreported warm-up execution on the first case. Formal CUDA timings use `time.perf_counter` with CUDA synchronization immediately before and after every timed case. Method order is rotated deterministically across the five repetitions to reduce order drift.

S5.2 Architecture-profiling protocol

The five executable architectures are profiled with batch size one on an Intel Core i7-12700H CPU and an NVIDIA RTX 3060 Laptop GPU for $H \in \{1, 4\}$. Each setting uses 100 warm-up inferences followed by five repetitions of 1000 timed inferences. CUDA execution is synchronized before and after each timed region. Median latency is the primary latency measure, and the 95th percentile describes measurement variability. Serialized parameter-file size is used as the measured model size. Peak allocated GPU memory is also recorded for each architecture–horizon pair. Table S5 reports the complete batch-one profile.

Table S5: Per-architecture batch-one hardware profile.

Configuration	H	Operations	Parameters	CPU (ms)	GPU (ms)	State (KiB)	GPU peak (KiB)
A1	1	155,712	77,921	0.097	0.148	306.5	1024
A1	4	155,904	78,020	0.094	0.133	306.9	1024
A6	1	157,760	78,977	0.093	0.168	311.0	1024
A6	4	157,952	79,076	0.099	0.151	311.4	1024
A13	1	159,808	80,033	0.128	0.237	315.5	1024
A13	4	160,000	80,132	0.128	0.218	315.9	1024
A55	1	377,888	2,081	1.793	3.999	10.3	54
A55	4	377,984	2,132	1.793	4.004	10.4	1030
A56/A57	1	1,050,688	5,697	1.980	4.019	24.4	60
A56/A57	4	1,050,880	5,796	1.975	4.170	24.8	1037

CPU and GPU values are median inference latency. A56 and A57 are the same executable one-layer GRU-32 architecture. GPU peak is the maximum allocated device memory recorded by the profiling protocol and is descriptive rather than a configured memory limit.

S5.3 Profiling results

Duplicate one-layer GRU-32 records are merged at the executable-architecture level. Across the CPU/GPU measurements and $H \in \{1, 4\}$, the Spearman correlation between estimated operation count and median inference latency ranges from 0.889 to 0.964. The correlation between parameter count and serialized parameter-file size is 1.000 in every measured setting. The MLP architectures use fewer estimated operations but more parameters than the GRU architectures. The two deployment indicators therefore describe different model properties, although this does not imply that both limits are active under every tested deployment setting. Peak GPU allocation is not monotonic in parameter count across the two horizons, so the parameter-count limit is interpreted as a model-size indicator rather than a direct device-memory limit. The measurements use one CPU/GPU host and are not a cross-platform deployment benchmark.

S5.4 Single-limit feasible-space audit

To determine which deployment limit is active in the frozen candidate bank, we apply the operation-count and parameter-count limits separately to the same seven candidates. This is a static complexity audit and does not repeat source training, target adaptation, model selection, or test evaluation.

For both $H = 1$ and $H = 4$, all seven candidates satisfy the operation-count limit under the tight, medium, and loose settings. Under the tight parameter-count limit, the three MLP candidates are excluded and the feasible bank is reduced from seven candidates to four, whereas the medium and loose parameter-count limits retain all seven candidates. Consequently, the joint feasible set contains four candidates under the tight setting and seven under the medium and loose settings. The fixed reference candidate satisfies both limits in all six horizon–tier combinations.

S5.5 Expanded-bank dual-limit stress audit

The preceding audit shows that the operation-count limit is inactive for the frozen seven-candidate main bank under the main limit tiers. To examine whether operation count and parameter count can independently restrict candidate feasibility, we additionally construct an eight-candidate stress bank by adding configuration 24 from the predefined architecture space.

The stress thresholds are determined only from deterministic model-complexity profiles, without using prediction-loss or target-performance information. Across $H \in \{1, 4\}$, the largest operation count in the original seven-candidate bank is 1,050,880, whereas the smallest operation count of configuration 24 is 1,198,112. Their midpoint gives the stress operation-count limit $B^F = 1{,}124{,}496$. Similarly, the largest parameter count among the stress-bank candidates other than A13 is 79,076, whereas the smallest parameter count of A13 is 80,033. Their midpoint gives the stress parameter-count limit $B^P = 79{,}554.5$.

For both $H = 1$ and $H = 4$, configuration 24 violates only the operation-count limit, A13 violates only the parameter-count limit, and the fixed reference A57 satisfies both limits. Each single-limit feasible set therefore retains seven of the eight candidates, whereas their joint feasible set retains six. Thus, the joint feasible set is a strict subset of each single-limit feasible set.

The target-side stress evaluation uses the same 80 held-out cases, and all outputs selected under the joint limits satisfy both limits. This stress audit does not modify the frozen seven-candidate main bank, the main deployment-limit tiers, or the headline experimental results.

S5.6 Selected-architecture analysis

Figure S2 summarizes the selected architectures, their complexity, and their paired performance.

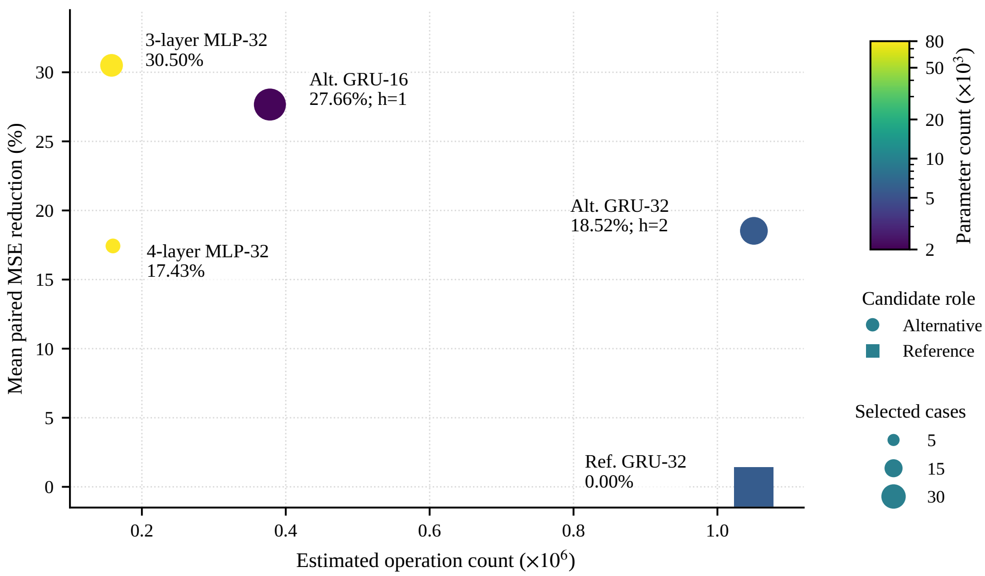

Figure S2: Selected-configuration complexity and performance. Marker area shows the number of selected cases, and h denotes harmful selections.

The reference configuration is retained in 33 cases. Four alternative configurations cover the remaining 47 cases, and their mean paired MSE reductions range from 17.43% to 30.50%. All four alternatives have positive mean reductions. The three harmful selections occur on two alternative GRU configurations.

S6 Configuration Analysis

Figure S3 reports the three configuration sensitivity analyses.

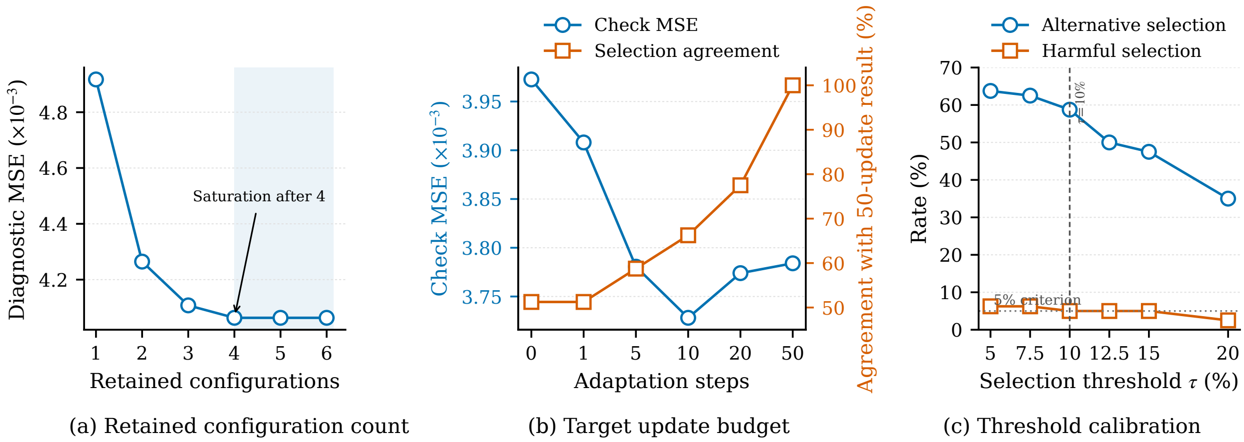

Figure S3: Configuration analysis of (a) the retained configuration count, (b) the target update budget, and (c) the relative margin.

Diagnostic MSE decreases from 0.004917 with one retained configuration to 0.004063 with four retained configurations and then remains unchanged. The complete retained list is kept because it was frozen before this diagnostic analysis. This result supports a compact candidate bank but does not identify a globally optimal bank size.

Most of the adaptation gain appears within the first ten target updates. The main experiment nevertheless uses 50 updates for every feasible candidate to preserve a common fixed target budget. The relative-margin analysis shows that 10% is the smallest value satisfying all preset calibration criteria.

S7 Source Sensitivity and External Transfer

S7.1 Source-center scale and source-training seeds

Figure S4 summarizes the source-scale, source-seed, and external-transfer results.

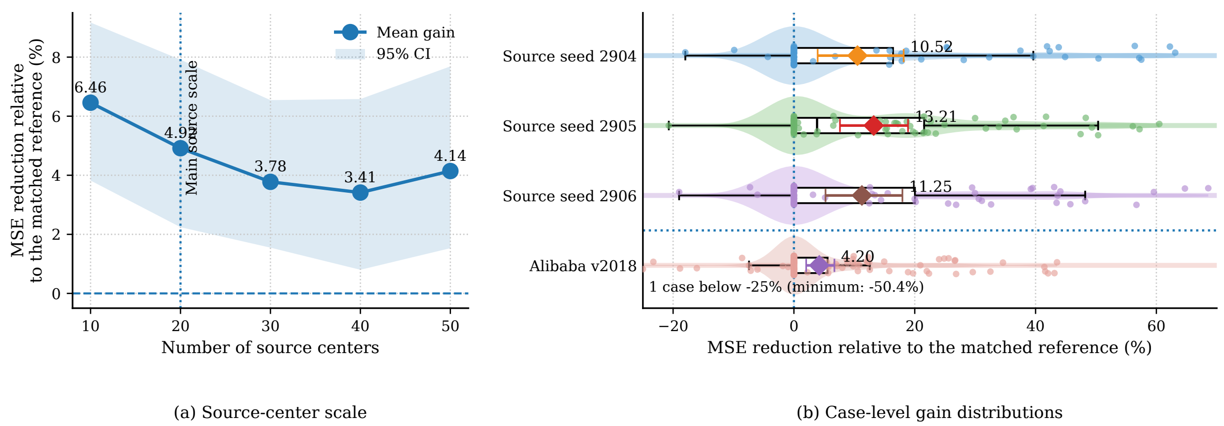

Figure S4: Sensitivity to source-center scale, source-training seeds, and Alibaba v2018 transfer.

One Alibaba case below −25% is clipped in Fig. S4 only for visualization; all original values are retained in the analysis.

For the source-scale analysis, scales 10, 20, 30, 40, and 50 are nested prefixes of the same 50-center source pool. Each architecture–horizon asset receives 2000 source updates with batch size 64, the same initialization across scales, and a random stream defined independently of scale. Every bank is then evaluated on the same disjoint 20-center pool (centers 1040–1059), with two horizons and two support sizes giving 80 cases. The architecture list, target update protocol, relative margin, target seeds, and evaluation pool are fixed. Because one controlled bank is trained at each scale, this is a descriptive scale-sensitivity analysis rather than an estimate over repeated source-center subsamples.

The mean paired MSE reductions at source-center scales 10, 20, 30, 40, and 50 are 6.46%, 4.92%, 3.78%, 3.41%, and 4.14%, respectively. The mean gain remains positive at every tested source scale.

Across source-training seeds 2904, 2905, and 2906, the mean paired MSE reductions are 10.52%, 13.21%, and 11.25%. All three confidence intervals remain above zero, although some individual cases have zero or negative gains. The screened architecture list is held fixed in this seed study. The result therefore measures sensitivity to source-model training and initialization. The three intervals are unadjusted pointwise 95% center-cluster intervals.

S7.2 Independent architecture-screening stability

We separately repeat the complete 66-configuration diagnostic screening on two new 20-center pools. Together with the original 20-center screening pool, this gives three independently sampled screening sets. The A57 reference, architecture space, initialization recipe, SGD/MSE–50 target recipe, and retention rule are unchanged. A configuration is retained when it records at least two check-oracle wins or at least two validation-selected positive check wins. The protected PT-A57 reference is not counted as retained; A57 in a retained list denotes the separately trained C1-A57 candidate. None of these pools overlaps the 20-center robustness pool used below, and Test is never read. Table S6 reports the retained lists and their evaluation on the untouched robustness pool.

Table S6: Independent candidate-screening stability and evaluation on centers 1160–1179.

Screen	Centers	Retained configurations	Size	Jaccard	Gain (%)	95% CI (%)	Harmful (%)
Frozen S0	900–919	A1, A6, A13, A55, A56, A57	6	1.000	0.295	$[-0.646, 1.564]$	2.50
Independent S1	1120–1139	reference fallback only	0	0.000	0.000	$[0.000, 0.000]$	0.00
Independent S2	1140–1159	A55, A56, A57, A59	4	0.429	0.125	$[-0.894, 1.427]$	3.75

Jaccard similarity is relative to the frozen S0 list. Size counts retained C1 configurations before adding the protected PT-A57 fallback; an A57 entry is therefore C1-A57, not PT-A57. Gain is the mean paired check-MSE reduction relative to PT-A57 on 80 robustness cases; intervals use a 4000-repetition center-cluster bootstrap. This is a C1 screening diagnostic and does not retrain or retune the frozen headline method.

Across S0–S2, retained-list sizes are 6, 0, and 4 (median 4, range 0–6). A55, A56, and A57 are retained in two of three pools; A1, A6, A13, and A59 are each retained once. On the robustness pool, the S0 list selects PT-A57 in 72 cases, A55 in three, A56 in three, and the C1 A57 initialization in two. S1 falls back to PT-A57 in all 80 cases. S2 selects PT-A57 in 71 cases, A55 in three, A56 in three, C1-A57 in two, and A59 in one.

The rule therefore does not recover a stable or unique architecture set, and neither nonempty list has a paired-gain interval excluding zero on the independent pool. This negative result narrows the claim: the main results evaluate a frozen, prespecified bank; they do not establish consistency of the preceding candidate-discovery rule. It also supports keeping the reference candidate outside the discovery decision and using it when no alternative satisfies the retention rule.

S7.3 Alibaba external-domain transfer

Alibaba Cluster Trace v2018 is split into 20 source machines, 20 calibration machines, and 40 held-out machines. The calibration and held-out pools contain 80 and 160 cases, respectively. Models are retrained using data from the Alibaba source machines and their own architectures and source-only normalization. The architecture definitions, deployment filtering rule, target protocol, and margin grid remain unchanged.

The workload observations are obtained from the production trace. The tight, medium, and loose deployment-limit tiers remain controlled model-complexity settings and are not direct measurements of machine-specific latency, memory use, or energy use. The Alibaba experiment therefore evaluates external workload-domain transfer under the frozen deployment protocol.

No margin in the fixed grid satisfies all calibration criteria on Alibaba. The previously frozen $\tau = 0.10$ rule is therefore evaluated without recalibration, and no Alibaba-specific calibrated margin is claimed.

On the 160 held-out cases, the adapted reference and MSA-DTI have mean MSE values of 0.001337 and 0.001279, respectively, corresponding to a 4.29% aggregate reduction. The mean paired MSE reduction is 4.20%, with a 95% confidence interval of $[2.03\%, 6.70\%]$. MSA-DTI selects alternatives in 57 cases, including 47 beneficial and 10 harmful selections. The harmful-selection rates are 6.25% over all held-out cases and 17.54% conditional on alternative selection. The all-case harmful-selection rate exceeds the original 5% criterion, so the external evaluation shows a positive average transfer gain but does not reproduce the original reliability criterion.

S8 Synthetic Benchmark Generation and Center Heterogeneity

The controlled benchmark represents heterogeneous computing centers in a common metric coordinate space. Each center may observe only a subset of the metric positions. The model input uses 12 value channels, 12 corresponding observation-mask channels, and one time-gap channel, giving a fixed input dimension of 25. Unavailable metric values are zero-filled and identified by the corresponding mask channels.

Center types A–C are evaluation strata rather than model classes. They differ in workload dynamics, sampling, and monitoring conditions. Deployment-limit tiers are separate platform-side settings and are assigned using the type-dependent probabilities reported in Table S7. The table also gives the complete type-specific parameter ranges. The default A:B:C ratio is 3:4:3, giving 6 Type-A, 8 Type-B, and 6 Type-C centers in each 20-center source or held-out target pool.

Table S7: Type-specific parameter ranges of the controlled synthetic benchmark.

Parameter	Type A	Type B	Type C
Sampling interval d_t	$\{1\}$	$\{1, 2\}$	$\{2, 3, 4\}$
Point-missing rate	$[0.02, 0.08]$	$[0.10, 0.25]$	$[0.25, 0.45]$
Missing-block length	$[5, 15]$	$[20, 60]$	$[60, 120]$
Observation-noise std.	$[0.01, 0.03]$	$[0.03, 0.06]$	$[0.06, 0.10]$
Permanently unavailable metrics	0–2	2–4	4–6
Base workload logit	$[-0.70, 0.10]$	$[-0.45, 0.35]$	$[-0.20, 0.55]$
Seasonal amplitude	$[0.20, 0.45]$	$[0.30, 0.60]$	$[0.40, 0.75]$
AR coefficient	$[0.55, 0.80]$	$[0.60, 0.88]$	$[0.65, 0.92]$
Process-noise std.	$[0.04, 0.08]$	$[0.06, 0.11]$	$[0.08, 0.14]$
Burst rate per 1000 base steps	$[1.0, 2.5]$	$[1.5, 3.5]$	$[2.0, 4.5]$
Burst amplitude	$[0.25, 0.55]$	$[0.35, 0.75]$	$[0.45, 0.95]$
Drift probability	0.20	0.40	0.60
Drift strength	$[0.10, 0.25]$	$[0.15, 0.35]$	$[0.20, 0.45]$
Limit-tier probabilities (tight / medium / loose)	0.25/0.45/0.30	0.35/0.45/0.20	0.45/0.40/0.15

The three types share the same missingness mechanisms. For every center, the generator may create 2–6 single-metric outage blocks and 1–3 grouped outage blocks containing 2–4 metrics. A short global monitoring outage occurs with probability 0.20. The shared periodic base scales are 24, 96, and 672 base steps.

Together, these settings vary five aspects of the benchmark: workload dynamics, sampling, monitored-variable availability, observation conditions, and deployment-limit settings.

For each center, the generator first creates a latent workload process. The workload contains periodic variation, autoregressive variation, and random workload bursts. When drift occurs, it is generated as either an abrupt change or a gradual transition.

The latent workload is then mapped to multivariate monitoring signals. Individual metrics may have different delays, gains, offsets, nonlinear terms, and observation noise. Center-specific sampling is applied before monitoring-schema and missingness effects are introduced.

The benchmark generator and all data-role assignments are fixed before architecture screening, margin calibration, diagnostics, and held-out evaluation.